Q P Code:-25900513

Mahatma Gandhi University, Kottayam

MGU-UGP (HONOURS) Regular Examination October 2025

Third Semester

Discipline Specific Elective (DSE)- MG3DSEMAT202 Numerical Methods

(2024 Admission Onwards)

Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A

(Short Answer Questions)
Answer any 5 questions
Each question carries 2 marks

Question 1 Find the relative error of the number 9.7 if both of its digits are correct. **[U] / [CO1]**

- For the arguments to get the maximum absolute error is $\Delta = 0.05$. ...**1 Mark**
- For writing The relative error = Absolute error/true value ...**1 Mark**
- relative error $= \dfrac{\Delta}{|9.7|} = \dfrac{0.05}{9.7} \approx 0.0051546 \approx 5.1546 \times 10^{-3}$. ...**1 Mark**

Maximum Marks should not exceed 2

Question 2 Compare the false position method and the bisection method. **[K] / [CO2]**

- For writing a brief explanation of false position method and the bisection method. ...**2 Marks**
- For writing their comparison ...**2 Marks**

Maximum Marks should not exceed 2

Question 3 Write the expression for the second forward difference $\Delta^2 y_i$. **[U] / [CO4]**

- For writing the first forward difference as $\Delta y_i = y_{i+1} - y_i$ ...**1 Mark**
- For taking the forward difference again, $\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i$ ...**1 Mark**
- Rest of the calculation to find $\Delta^2 y_i = y_{i+2} - 2y_{i+1} + y_i$ ...**1 Mark**

Maximum Marks should not exceed 2

Question 4 Define the differential operator D. **[K] / [CO4]**

- The differential operator D acts on sufficiently smooth functions as

$$D\{f(x)\} = \frac{d}{dx}f(x).$$

. . . 2 Marks

Question 5 Give the equation of a Lagrange polynomial of degree two passing through (x_0, y_0), (x_1, y_1) and (x_2, y_2). **[U] / [CO3]**

- The quadratic Lagrange interpolating polynomial is
$P_2(x) = y_0\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + y_1\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + y_2\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}.$

. . . 2 Marks

Question 6 Write first approximation formula in the method of successive approximations. **[K] / [CO4]**

- Given an equation $x = g(x)$, the iteration (method of successive approximations / fixed-point iteration) is $x_{n+1} = g(x_n)$.6 A first approximation uses an initial guess x_0 and computes $x_1 = g(x_0)$ **. . . 2 Marks**

Question 7 Using the following data, find $\frac{dy}{dx}$ at $x = 7$ **[A] / [CO4]**

x	1	3	5	7
y	6	11	19	27

- For constructing the forward/b6ackward difference table. **. . . 2 Marks**
- For using Newton's backward difference formula for the derivative at the last point $x_n = 7$ **. . . 2 Marks**
- Getting $\left(\frac{dy}{dx}\right)_{x=7} = 3.5$ **. . . 2 Marks**

Maximum Marks should not exceed 2

Question 8 Using Stirling's formula, derive the expression for the first derivative and second derivative of a function at a tabular point. **[A] / [CO4]**

- For writing Stirling's central difference interpolation formula about the central value x_0 is

$$f(x) = f_0 + p\frac{\Delta f_0 + \Delta f_{-1}}{2} + \frac{p^2}{2!}\Delta^2 f_{-1} + \frac{p(p^2-1)}{3!}\frac{\Delta^3 f_{-1} + \Delta^3 f_{-2}}{2} + \cdots$$

where $p = \frac{x - x_0}{h}$. **. . . 2 Marks**

- For finding First derivative:

$$\left(\frac{df}{dx}\right)_{x_0} = \frac{1}{h}\left[\frac{\Delta f_0 + \Delta f_{-1}}{2} - \frac{1}{12}(\Delta^3 f_{-1} + \Delta^3 f_{-2}) + \cdots\right]$$

. . . 2 Marks

- For finding the Second derivative:

$$\left(\frac{d^2 f}{dx^2}\right)_{x_0} = \frac{1}{h^2}\left[\Delta^2 f_{-1} - \frac{1}{12}(\Delta^4 f_{-2} + \Delta^4 f_{-3}) + \cdots\right].$$

. . . 2 Marks

Maximum Marks should not exceed 2

Part B
(Short Essay Questions)
Answer any 5 questions
Each question carries 6 marks

Question 9 Find a root of the equation $x^3 - 4x - 9 = 0$ in the interval $[2, 3]$ by the bisection method. Perform 5 iterations. **[A]/[CO2]**

- For finding $f(2)$ and $f(3)$ have opposite signs and writing a root exists in the interval $[2, 3]$. **. . . 2 Marks**
- For the first iteration **. . . 2 Marks**
- Rest of the four iterations and the final answer $x \approx 2.71875$ **. . . 3 Marks**

Maximum Marks should not exceed 6

Question 10 Solve $f(x) = x^2 - 7 = 0$ using Newton–Raphson method (two iterations). **[A]/[CO2]**

- For taking $f(x) = x^2 - 7$, $f'(x) = 2x$ and choose $x_0 = 3$ as the initial guess **. . . 2 Marks**
- For writing the iteration formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

. . . 2 Marks

- For the first iteration **. . . 2 Marks**
- Rest of the calculations and final answer $x \approx 2.6458$ **. . . 1 Mark**

Maximum Marks should not exceed 6

Question 11 Choosing Newton's backward difference interpolation, fit a polynomial for the data $(1, 15), (3, 27)$ and $(5, 41)$. **[A]/[CO3]**

- For writing Newton's backward interpolation formula

$$y(x) = y_n + p\,\nabla y_n + \frac{p(p+1)}{2!}\,\nabla^2 y_n + \cdots, \quad \text{where} \quad p = \frac{x - x_n}{h} = \frac{x-5}{2}.$$

. . . 2 Marks

- For getting $y(x) = 41 + 14p + p(p+1)$. **. . . 2 Marks**
- For substituting $p = \dfrac{x-5}{2}$ and simplifying **. . . 2 Marks**
- Calculations to find $y(x) = 41 + (7x - 35) + \dfrac{(x-5)(x-3)}{4}$ **. . . 2 Marks**

Maximum Marks should not exceed 6

Question 12 Evaluate: **[A]/[CO4]**

(a) $\Delta^2(x^3 + 3x + 5)$

- Take $h = 1$. For writing $\Delta f(x) = f(x+1) - f(x)$ and $\Delta^2 f(x) = f(x+2) - 2f(x+1) + f(x)$. **. . . 2 Marks**
- Getting $f(x+1) = x^3 + 3x^2 + 6x + 9$ and $f(x+2) = x^3 + 6x^2 + 15x + 19$. **. . . 2 Marks**
- Concluding that $\Delta^2(x^3 + 3x + 5) = 6x + 6$. **. . . 2 Marks**

(b) $\Delta(x^3 + 5x^2 - 2x - 10)$

- Getting $f(x+1) = x^3 + 8x^2 + 11x - 6$. **. . . 2 Marks**
- For calculating $\Delta(x^3 + 5x^2 - 2x - 10) = 3x^2 + 13x + 4$. **. . . 2 Marks**

N.B:- "Answers in which h appears unchanged are eligible for full marks."
Maximum Marks should not exceed 6

Question 13 Determine the polynomial $f(x)$ that fits the following data using Newton's divided differences: **[U]/[CO3]**

x	0	2	3	5
$f(x)$	1	5	10	26

- For computing divided differences. **. . . 4 Marks**
- For writing Newton's divided difference interpolation polynomial

$$P_n(x) = f[x_0] + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \cdots + (x - x_0)(x - x_1)\cdots(x - x_{n-1})f[x_0, x_1, \ldots, x_n].$$

. . . 2 Marks

- For obtaining $P(x) = x^2 + 1$ ... **2 Marks**

Maximum Marks should not exceed 6

Question 14 From the following table, estimate the value of x corresponding to $y = 6$. **[U]/[CO4]**

y	2	4	8
x	1	2	3

- For writing Lagrange's interpolation formula is given by:

$$x = x_0\frac{(y-y_1)(y-y_2)}{(y_0-y_1)(y_0-y_2)} + x_1\frac{(y-y_0)(y-y_2)}{(y_1-y_0)(y_1-y_2)} + x_2\frac{(y-y_0)(y-y_1)}{(y_2-y_0)(y_2-y_1)}.$$

... **3 Marks**

- **Compute for** $y = 6$

$$x = 1\cdot\frac{(6-4)(6-8)}{(2-4)(2-8)} + 2\cdot\frac{(6-2)(6-8)}{(4-2)(4-8)} + 3\cdot\frac{(6-2)(6-4)}{(8-2)(8-4)}$$

$$= 2.67$$

... **4 Marks**

Maximum Marks should not exceed 6

Question 15 The distance (x cm) traversed by a particle at different times (t seconds) are given below:

t	0.0	0.1	0.2	0.3	0.4	0.5	0.6
x	3.01	3.16	3.29	3.36	3.40	3.38	3.32

Find the acceleration of the particle and estimate the error at $t = 0.3$ sec. **[An]/[CO1]**

- For writing Stirling's Central Difference Formula

$$f(x) = f_0 + p\frac{\Delta f_0 + \Delta f_{-1}}{2} + \frac{p^2}{2!}\Delta^2 f_{-1} + \frac{p(p^2-1)}{3!}\frac{\Delta^3 f_{-1} + \Delta^3 f_{-2}}{2} + \frac{p^2(p^2-1)}{4!}\Delta^4 f_{-2} + \cdots$$

where $p = \dfrac{x - x_0}{h}$. ... **3 Marks**

- Obtaining $\left(\dfrac{d^2f}{dx^2}\right)_{x_0}$ at the central point $x = x_0$

$$\left(\frac{d^2f}{dx^2}\right)_{x_0} = \frac{1}{h^2}\left[\Delta^2 f_{-1} - \frac{1}{12}\Delta^4 f_{-2} + \frac{1}{90}\Delta^6 f_{-3} - \cdots\right]$$

... **3 Marks**

- Rest of the calculation to get $a(0.3) \approx -3.0\text{cm/s}^2$. ... **2 Marks**
- Obtaining estimate the of the error 0.5 cm/s^2 ... **2 Marks**

Maximum Marks should not exceed 6

Question 16 Using the following data, find the area bounded by the curve $f(x)$ and the x-axis from $x = 7.47$ to $x = 7.52$.

x	7.47	7.48	7.49	7.50	7.51	7.52
$f(x)$	1.93	1.95	1.98	2.01	2.03	2.06

[A]/[CO4]

- Writing the formula for the composite trapezoidal rule is:

$$\int_a^b f(x)\,dx \approx h\left[\frac{f_0 + f_n}{2} + \sum_{i=1}^{n-1} f_i\right]$$

... **3 Marks**

- Getting

$$h = 7.48 - 7.47 = 0.01$$

.

$$\sum_{i=1}^{4} f_i = 1.95 + 1.98 + 2.01 + 2.03 = 7.97,$$

$$\frac{f_0 + f_5}{2} = \frac{1.93 + 2.06}{2} = \frac{3.99}{2} = 1.995.$$

... **3 Marks**

- For Substituting in the formula,

$$\int_{7.47}^{7.52} f(x)\,dx \approx 0.01\,[1.995 + 7.97]$$

$$= 0.01 \times 9.965$$

$$= 0.09965.$$

... **2 Marks**

Maximum Marks should not exceed 6

Part C
(Essay Questions)
Answer any 3 questions
Each question carries 10 marks

Question 17 Solve $x^3 - 2x - 5 = 0$ using iteration method up to five iterations. **[E]/[CO2]**

- For taking $x = g(x) = (2x + 5)^{1/3}$. ... **4 Marks**
- For the first iteration ... **4 Marks**
- Rest of the four iterations and final answer $x \approx 2.09453$ ... **4 Marks**

Maximum Marks should not exceed 10

Question 18 Find the number of men getting wages between 10 and 15 rupees by using Newton's Forward Difference Interpolation from: **[A]/[CO3]**

Wages	0−10	10−20	20−30	30−40
No. of Men	9	30	35	42

- For taking data points from the table as

x	10	20	30	40
y	9	39	74	116

... **3 Marks**

- Writing Newton's forward interpolation formula as

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

where

$$p = \frac{x - x_0}{h}.$$

... **3 Marks**

- For obtainingThe forward difference table ... **3 Marks**
- For finding the number of men getting wages between 10 and 15 is 14.5.

... **3 Marks**

Maximum Marks should not exceed 10

Question 19 Determine y when $x = 3.2$ from the data $y(2.0) = 246.2, y(2.5) = 409.3, y(3.0 = 537.2, y(3.5) = 636.3$ and $y(4.0) = 715.9$ using (i) Gauss's forward formula and (ii) Gauss's backward formula. **[A]/[CO4]**

- For taking the table as

x	2.0	2.5	3.0	3.5	4.0
y	246.2	409.3	537.2	636.3	715.9

... **3 Marks**

- For computing forward differences Δ, Δ^2, Δ^3, Δ^4 ... **4 Marks**

- For writing Gauss's forward formula about a central point x_0 as

$$y(x) \approx y_0 + p\,\Delta y_0 + \frac{p(p-1)}{2!}\,\Delta^2 y_{-1} + \frac{p(p-1)(p+1)}{3!}\,\Delta^3 y_{-1} + \frac{p(p-1)(p+1)(p-2)}{4!}\,\Delta^4 y_{-2} + \cdots$$

. . . 3 Marks

- For the calculations with Gauss's forward formula to get $y(3.2) \approx 580.18512$

. . . 2 Marks

- For writing backward formula

$$y(x) \approx y_0 + p\,\Delta y_{-1} + \frac{p(p+1)}{2!}\,\Delta^2 y_{-1} + \frac{p(p+1)(p-1)}{3!}\,\Delta^3 y_{-1} + \frac{p(p+1)(p-1)(p+2)}{4!}\,\Delta^4 y_{-2} + \cdots$$

. . . 2 Marks

- For the calculations with Gauss's backward formula to get $y(3.2) \approx 580.18512$

. . . 2 Marks

Maximum Marks should not exceed 10

Question 20 Evaluate $f(0.23)$ and $f(0.29)$ from the following data by Lagrange's Interpolation: **[E]/[CO3]**

x	0.20	0.22	0.24	0.26	0.28	0.30
$f(x)$	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

- For writing the general form of the **Lagrange interpolation formula** is:

$$f(x) = \sum_{i=0}^{n} f(x_i) \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

. . . 4 Marks

- For calculations to the Evaluation of $f(0.23) = 1.6751$ **. . . 4 Marks**
- For calculations to the Evaluation of $f(0.29) = 1.7081$ **. . . 4 Marks**

Question 21 Evaluate $\displaystyle\int_0^{0.6} e^{-x^2}\,dx$ using Simpson's $\frac{1}{3}$ rule with $h = 0.1$. **[A]/[CO4]**

- For taking Simpson $\frac{1}{3}$ rule :

$$I \approx \frac{h}{3}\left[f_0 + f_6 + 4\sum_{i \text{ odd}} f_i + 2\sum_{i \text{ even},\ 2 \leq i \leq 4} f_i\right].$$

Step size:

$$h = 0.1$$

Number of subintervals: $n = \dfrac{0.6 - 0}{0.1} = 6$ **. . . 4 Marks**

- For obtaining the table as

x_i	0.0	0.1	0.2	0.3	0.4	0.5	0.6
$f(x_i) = e^{-x_i^2}$	1.0000	0.9900	0.9607	0.9139	0.8521	0.7788	0.6976

. . . 4 Marks

- By the rule $\int_0^{0.6} e^{-x^2}\,dx = 0.53515$ **. . . 4 Marks**

Maximum Marks should not exceed 10

Question 22 Approximate $\int_0^2 e^x\,dx$ using Simpson's $\frac{3}{8}$ rule with 6 subintervals. **[A]/[CO4]**

- For writing **Simpson's $\frac{3}{8}$ Rule:**

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f(x_0) + f(x_n) + 3\sum_{\substack{1\le i\le n-1\\ i\not\equiv 0 \pmod 3}} f(x_i) + 2\sum_{\substack{1\le i\le n-1\\ i\equiv 0 \pmod 3}} f(x_i)\right]$$

. . . 3 Marks

- Taking **Interval:** $[0, 2]$

$$n = 6, \qquad h = \frac{2-0}{6} = \frac{1}{3} \approx 0.333$$

and the table as

x_i	0	0.333	0.666	1.000	1.333	1.666	2.000
$f(x_i) = e^{x_i}$	1.0000	1.3956	1.9477	2.7182	3.7936	5.2944	7.3890

. . . 4 Marks

- By the rule $\int_0^2 e^x\,dx = 6.390$ **. . . 4 Marks**

Maximum Marks should not exceed 10

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.